

Consumption

Macroeconomics B

Lecture 3 (Chapter 16)

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Where We Are

In Lecture 1 you studied **investment and asset prices** in Chapter 15:

- ▶ Firms invest until the shadow value of capital equals the marginal installation cost, $q_t = 1 + c'(I_t)$.
- ▶ Investment depends positively on q and expected profitability, and negatively on the real interest rate.
- ▶ Asset prices reflect discounted future payoffs.

Today you move to **Chapter 16**, the other major component of private aggregate demand. Consumption accounts for roughly 60–70 % of GDP in most economies.

Key Question: does consumption depend mainly on *current* income, or on *lifetime* resources? The answer determines how effective fiscal policy can be.

Roadmap for Today

1. **The Keynesian benchmark and the empirical puzzle**
2. **Intertemporal optimization: IBC and Keynes-Ramsey rule**
3. **The consumption function and the permanent income hypothesis**
4. **Ricardian equivalence: taxes, debt, and consumption**
5. **Credit constraints and precautionary saving**
6. **What the data say: MPC heterogeneity**
 - *Notebook exercise*

What You Will Be Able to Do After This Lecture

After this lecture, **you** will be able to:

- ▶ explain why the Keynesian consumption function cannot simultaneously fit cross-section and time-series evidence,
- ▶ derive the intertemporal budget constraint and interpret human wealth,
- ▶ apply the Keynes-Ramsey rule to determine the optimal consumption path and relate it to the interest rate and patience,
- ▶ state the permanent income hypothesis and compute the marginal propensity to consume out of a temporary versus a permanent income change,
- ▶ state Ricardian equivalence and identify at least three conditions under which it breaks down in practice,
- ▶ explain how precautionary saving and borrowing constraints create a stronger link between consumption and current income than the PIH predicts.

Setting the Stage: Fixed-Rate and Adjustable-Rate Mortgages

Before turning to the model, we need one institutional distinction.

- ▶ A **fixed-rate mortgage** keeps the interest rate fixed for a long period. Monthly payments are therefore relatively stable as long as the household does not refinance.
- ▶ An **adjustable-rate mortgage (ARM)** has a short interest-rate fixation period. When market rates rise and the loan resets, monthly payments rise as well.
- ▶ This matters for consumption because the **same monetary shock** can create very different cash-flow exposure across otherwise similar households.

Source: Danmarks Nationalbank, *Financial Stability, 1st Half 2022*; Danmarks Nationalbank, *The Household Cash-Flow Effects of Monetary Policy in Denmark and the Euro Area*, Economic Memo, 8 January 2024.

Motivation: Why Did Some Households Cut Spending More in 2022–23?

From 2022, the ECB and Danmarks Nationalbank raised policy rates sharply after many years of very low rates.

- ▶ Danmarks Nationalbank stresses that this tightening can have different **cash-flow effects** across households, even when the aggregate shock is the same.
- ▶ In Denmark, this heterogeneity is especially visible because mortgage contracts differ in how quickly higher market rates pass through to monthly payments.
- ▶ The Danish case is therefore a clean illustration of a general macroeconomic point: consumption reacts not only to income, but also to wealth, interest rates, and financing frictions.

Three Channels We Want to Separate:

1. **Intertemporal substitution:** a higher real interest rate makes current consumption relatively more expensive.
2. **Wealth effects:** higher discount rates and lower house prices reduce total wealth.
3. **Cash-flow and borrowing constraints:** some households must cut spending because financing conditions tighten immediately.

Today's consumption model gives you the language to distinguish these channels.

The Keynesian Consumption Function

Keynes (1936) proposed that consumption depends on *current* disposable income:

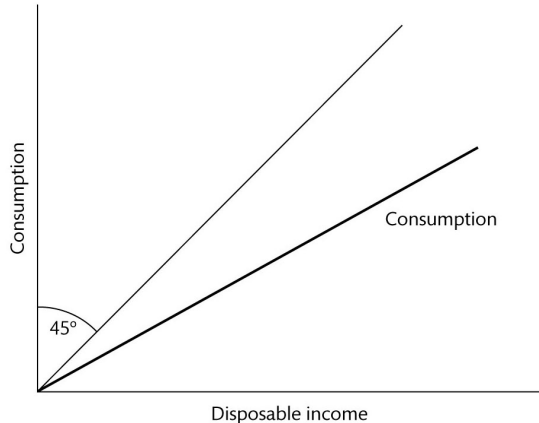
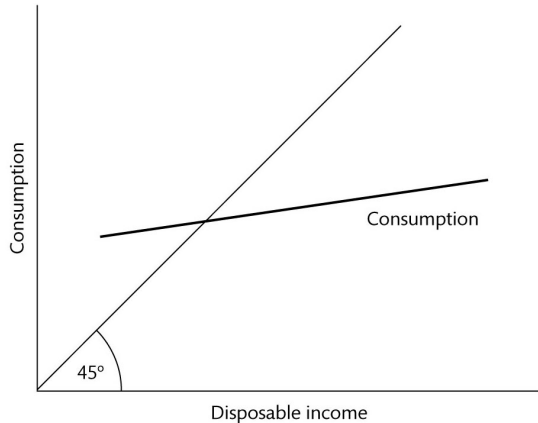
$$C_t = \bar{C} + b Y_t^d \quad \bar{C} > 0, \quad 0 < b < 1.$$

- ▶ $b \equiv MPC$ is the **marginal propensity to consume**.
- ▶ The **average propensity to consume**: $APC_t \equiv C_t/Y_t^d = b + \bar{C}/Y_t^d$ is *decreasing* in income.
- ▶ The psychological law: people raise spending as income rises, but by less than one-for-one.
- ▶ Policy: a transfer of ΔY^d raises aggregate demand by $b \cdot \Delta Y^d$ immediately.

Problem: two empirical patterns that this function cannot simultaneously explain.

Historical note

Two Contradictory Patterns

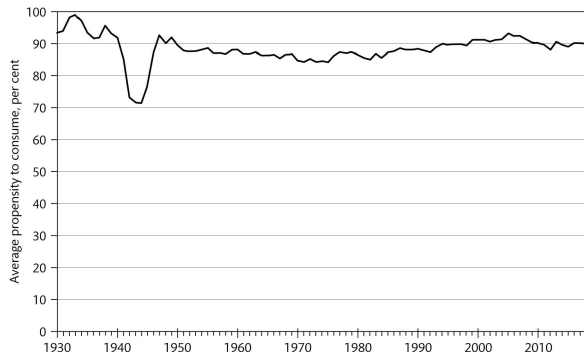


Cross-section: richer households have a lower APC.
Keynesian function fits.

Time-series: aggregate consumption and income grow together. APC appears stable.

Source: Sørensen & Whitta-Jacobsen, Figures 16.1a and 16.1b.

The Empirical Puzzle



Source: Sørensen & Whitta-Jacobsen, Figure 16.2. US average propensity to consume, 1929–2010 (apart from WWII rationing).

The aggregate APC in the US has been *remarkably stable* over 80 years, despite massive income growth.

The Keynesian function predicts APC should *fall* as income grows. The data say otherwise.

We need a theory in which:

- ▶ APC declines with income *cross-sectionally*,
- ▶ but is roughly constant *in the aggregate* over time.

The Two-Period Setup

We model a household planning over **two periods** ($t = 1, 2$).

- ▶ C_1, C_2 : real consumption in each period.
- ▶ Y_1^L, Y_2^L : pre-tax labour income.
- ▶ T_1, T_2 : lump-sum taxes net of transfers.
- ▶ Disposable income:

$$Y_t^d \equiv Y_t^L - T_t.$$

- ▶ V_1 : initial real financial wealth.
- ▶ A_1 : assets carried from period 1 into period 2.
- ▶ A_2 : assets left at the end of period 2.
- ▶ r : real interest rate.

With perfect capital markets, the household can borrow or save at rate r .

From Period Budgets to the Lifetime Budget

The period budget constraints are

$$\begin{aligned}C_1 + A_1 &= V_1 + Y_1^d, \\C_2 + A_2 &= (1 + r)A_1 + Y_2^d.\end{aligned}$$

The household cannot end life with unpaid debt:

$$A_2 \geq 0.$$

Now substitute the first constraint into the second:

$$A_1 = V_1 + Y_1^d - C_1,$$

so

$$C_2 + A_2 = (1 + r)(V_1 + Y_1^d - C_1) + Y_2^d.$$

Rearranging gives

$$C_1 + \frac{C_2}{1 + r} = V_1 + Y_1^d + \frac{Y_2^d}{1 + r} - \frac{A_2}{1 + r}.$$

The Intertemporal Budget Constraint

Since utility is increasing in consumption, the household does not leave valuable resources unused in the final period. Hence, in optimum, $A_2 = 0$.

The lifetime budget constraint becomes

$$C_1 + \frac{C_2}{1+r} = V_1 + Y_1^d + \frac{Y_2^d}{1+r}$$

Define **human wealth** as the present value of after-tax labour income:

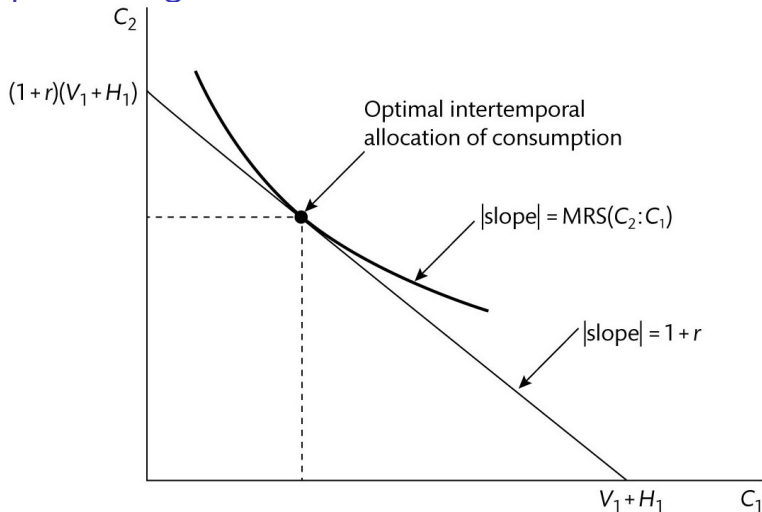
$$H_1 \equiv Y_1^d + \frac{Y_2^d}{1+r} = (Y_1^L - T_1) + \frac{Y_2^L - T_2}{1+r}.$$

Thus:

$$C_1 + \frac{C_2}{1+r} = V_1 + H_1$$

- ▶ Left side: present value of lifetime consumption.
- ▶ Right side: total initial wealth, financial wealth plus human wealth.
- ▶ Capital markets allow households to separate the timing of income from the timing of consumption.

The Intertemporal Budget Line



Source: Sørensen & Whitta-Jacobsen, Figure 16.3. The budget line has slope $-(1+r)$ and passes through $(V_1 + H_1, (1+r)(V_1 + H_1))$.

A higher r rotates the budget line and changes the relative price of current consumption.

Preferences: Lifetime Utility

The household maximises a **lifetime utility function**:

$$U = u(C_1) + \frac{1}{1+\phi} u(C_2), \quad u'(C) > 0, \quad u''(C) < 0, \quad \phi > 0.$$

- ▶ $\phi > 0$: the **rate of time preference**. A positive ϕ means the household is impatient: a unit of utility today is valued more than a unit tomorrow.
- ▶ $u'' < 0$: diminishing marginal utility. The household dislikes volatility in consumption across periods.
- ▶ Implication: if ϕ is not too large, the household will spread consumption evenly over time rather than concentrating it in one period.

The household chooses C_1 (and implicitly C_2 via the IBC) to maximise U .

From the IBC to the Optimal Consumption Choice

The household chooses C_1 and C_2 to maximise

$$U = u(C_1) + \frac{1}{1 + \phi} u(C_2)$$

subject to

$$C_1 + \frac{C_2}{1 + r} = V_1 + H_1.$$

- ▶ If you increase C_1 by 1 unit, you must reduce C_2 by $1 + r$ units to stay on the budget line.
- ▶ So the choice is a **marginal trade-off**: more utility today, less discounted utility tomorrow.
- ▶ At the optimum, the household stops reallocating consumption across time because the gain from shifting one more unit is zero.

This logic gives the Keynes-Ramsey rule.

Deriving the Keynes-Ramsey Rule Step by Step

Use the IBC to write future consumption as

$$C_2 = (1 + r)(V_1 + H_1 - C_1).$$

Insert this into lifetime utility:

$$U(C_1) = u(C_1) + \frac{1}{1 + \phi} u((1 + r)(V_1 + H_1 - C_1)).$$

Differentiate with respect to C_1 :

$$\frac{dU}{dC_1} = u'(C_1) - \frac{1 + r}{1 + \phi} u'(C_2).$$

Set the derivative equal to zero:

$$u'(C_1) = \frac{1 + r}{1 + \phi} u'(C_2)$$

Mechanism: the marginal utility loss from cutting C_1 by 1 must match the discounted marginal utility gain from increasing C_2 by $1 + r$.

Economic Meaning of the Keynes-Ramsey Rule

$$u'(C_1) = \frac{1+r}{1+\phi} u'(C_2)$$

- ▶ **Left side:** marginal utility from consuming one more unit today.
- ▶ **Right side:** discounted utility value of saving that unit and consuming $(1+r)$ more units tomorrow.
- ▶ If r rises, postponing consumption becomes more attractive, because one unit saved today buys more tomorrow.
- ▶ If ϕ rises, the household becomes more impatient, so current consumption becomes relatively more attractive.

Special cases:

- ▶ If $r = \phi$, then $u'(C_1) = u'(C_2)$ and concavity implies $C_1 = C_2$: perfect smoothing.
- ▶ If $r > \phi$, the return to waiting is strong, so the household tilts consumption towards period 2.
- ▶ If $r < \phi$, impatience dominates, so the household tilts consumption towards period 1.

Consumption Smoothing: Two Cases

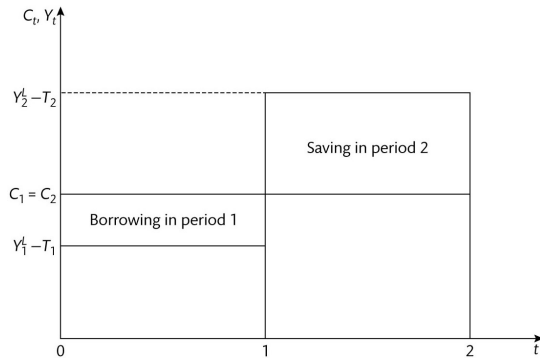


Fig. 16.4a: low income in period 1. The household borrows to smooth consumption over time.

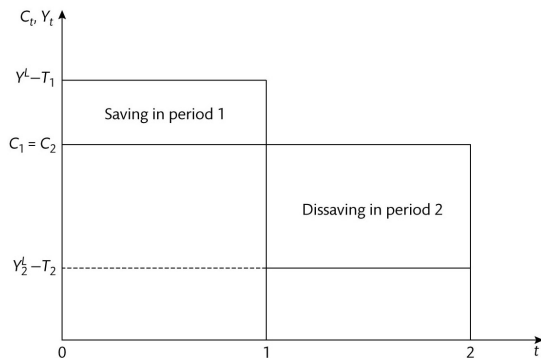


Fig. 16.4b: high income in period 1. The household saves to smooth consumption over time.

Source: Sørensen & Whitta-Jacobsen, Figures 16.4a–16.4b. Both panels: $r = \phi$, $V_1 = 0$, so the household chooses $C_1 = C_2$. Consumption reacts much less than income because the household spreads the shock over both periods.

Closed-Form Consumption Function with CRRA Utility

Assume CRRA preferences:

$$u(C) = \frac{C^{1-1/\sigma}}{1-1/\sigma}, \quad u'(C) = C^{-1/\sigma},$$

where $\sigma > 0$ is the **intertemporal elasticity of substitution**. Then the Keynes-Ramsey rule becomes

$$C_1^{-1/\sigma} = \frac{1+r}{1+\phi} C_2^{-1/\sigma} \Rightarrow C_2 = \left(\frac{1+r}{1+\phi} \right)^\sigma C_1.$$

- ▶ This tells you how the household wants to **tilt** consumption across time.
- ▶ A higher r raises desired future consumption relative to current consumption.
- ▶ A higher σ means the household is more willing to substitute across periods, so the tilt reacts more strongly to r .

Solving for Current Consumption

Insert

$$C_2 = \left(\frac{1+r}{1+\phi} \right)^\sigma C_1$$

into the IBC:

$$C_1 + \frac{1}{1+r} \left(\frac{1+r}{1+\phi} \right)^\sigma C_1 = V_1 + H_1.$$

Factor out C_1 :

$$C_1 [1 + (1+\phi)^{-\sigma}(1+r)^{\sigma-1}] = V_1 + H_1.$$

So the consumption function is

$$\boxed{C_1 = \theta(V_1 + H_1)} \quad \theta = \frac{1}{1 + (1+\phi)^{-\sigma}(1+r)^{\sigma-1}}.$$

Mechanism: current consumption is a fixed share of total wealth, but that share depends on impatience, the interest rate, and the willingness to substitute across time.

What Drives Current Consumption?

From $C_1 = \theta(V_1 + H_1)$, current consumption rises with:

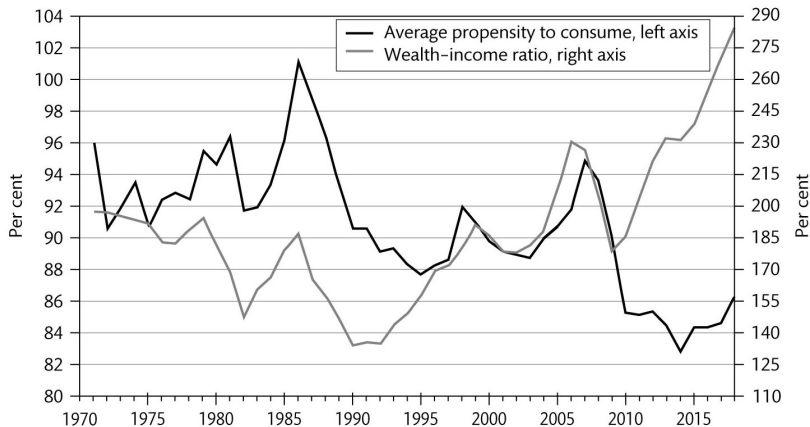
- ▶ V_1^+ : financial wealth. Stock market gains, inheritance, or rising house prices feed directly into C_1 .
- ▶ Y_1^d : current disposable income (part of H_1).
- ▶ $(Y_2^d)^e$: expected future income (the other part of H_1). A student expecting a high starting salary may consume more today.

The **interest rate effect** combines three channels:

- ▶ *Substitution effect*: $r \uparrow \Rightarrow C_1 \downarrow$ (current consumption becomes relatively more expensive).
- ▶ *Income effect*: $r \uparrow \Rightarrow C_1 \uparrow$ for a net saver (more interest income).
- ▶ *Wealth effect*: $r \uparrow \Rightarrow H_1 \downarrow$ and financial asset prices fall, reducing V_1 .

Empirically, the wealth effect tends to dominate, so $r \uparrow$ reduces consumption on net.

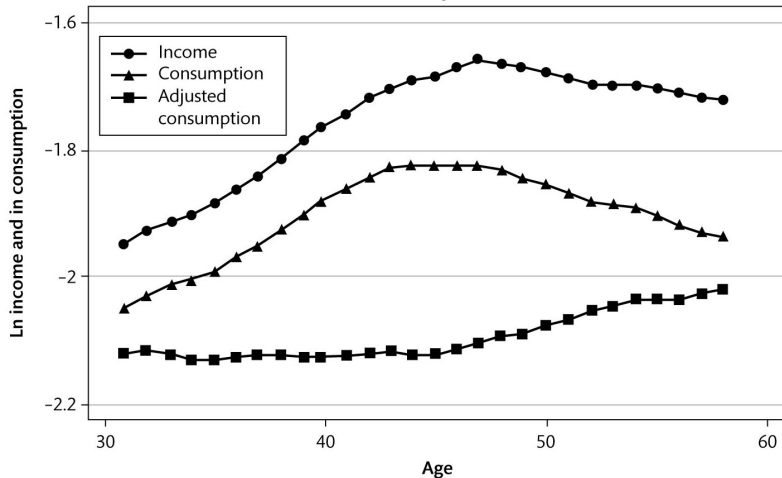
Danish Evidence: Consumption and Wealth



Source: Sørensen & Whitta-Jacobsen, Figure 16.7. Propensity to consume and wealth-income ratio in Denmark, 1971–2018 (Danish Economic Council; SMEC database).

A higher wealth-income ratio is associated with a higher propensity to consume, consistent with the wealth channel.

Life-Cycle Profiles: Income and Consumption over the Lifetime



Source: Sørensen & Whitta-Jacobsen, Figure 16.5. Consumption and income profiles for UK households born 1935–1939. “Adjusted consumption” corrects for the changing number of children.

Consumption is much smoother than income over the life cycle.

The Permanent Income Hypothesis

With many periods, current consumption is linked to **total lifetime wealth**, not just current income. Let

$$w_t = V_t + H_t$$

be financial wealth plus human wealth. If $r = \phi$, the household wants to smooth consumption over time, so current consumption equals the **annuity value** of total wealth:

$$C_t = y_t^P = \frac{r}{1+r} w_t$$

where y_t^P is **permanent income**.

- ▶ **What we learn:** households do not ask, “How much did my income move today?” They ask, “How much did my lifetime resources move?”
- ▶ A one-off income change has only a small effect on lifetime wealth, so consumption moves little.
- ▶ A persistent income change raises lifetime wealth much more, so consumption moves much more.
- ▶ This is why the PIH can explain a stable aggregate APC even when current income grows over time.

From Lifetime Wealth to an Annuity Rule

Suppose the household plans over the next N periods and chooses a **flat** consumption path, $C_t = C_{t+1} = \dots = C_{t+N-1}$, which is optimal when $r = \phi$. Then the present-value budget constraint is

$$\sum_{j=0}^{N-1} \frac{C_t}{(1+r)^j} = W_t,$$

where W_t is total wealth at time t . Factor out C_t :

$$C_t \sum_{j=0}^{N-1} (1+r)^{-j} = W_t.$$

So

$$C_t = \theta_N W_t, \quad \theta_N = \left(\sum_{j=0}^{N-1} (1+r)^{-j} \right)^{-1}$$

Mechanism: current consumption is the constant annuity that can be financed by lifetime wealth. A larger horizon means each one-off gain is spread more thinly across time.

Evaluating the Annuity Factor

From $C_t \sum_{j=0}^{N-1} (1+r)^{-j} = W_t$, we need to evaluate the present-value factor explicitly. It is a finite geometric series:

$$\sum_{j=0}^{N-1} (1+r)^{-j} = \frac{1 - (1+r)^{-N}}{1 - (1+r)^{-1}} = \frac{1+r}{r} \left[1 - (1+r)^{-N} \right].$$

Inverting gives the marginal propensity to consume out of total wealth:

$$\theta_N = \frac{r}{1+r} \cdot \frac{1}{1 - (1+r)^{-N}}$$

- ▶ For large N , $(1+r)^{-N} \rightarrow 0$, so $\theta_N \rightarrow r/(1+r)$: the household spends roughly the interest earned on its wealth each period.
- ▶ For small N (few periods left), θ_N is large: the household must run down wealth quickly and consumes a larger share each period.
- ▶ **Implication:** a one-off windfall has an extremely small effect on consumption for a young household with many periods ahead.

Temporary Versus Permanent Income: Step by Step

Start from

$$C_t = \theta_N W_t.$$

Now compare two income changes.

1. Temporary income increase of 1 today

$$dW_t = 1 \quad \Rightarrow \quad \boxed{dC_t = \theta_N}$$

Only current wealth rises by 1, so the extra resources are spread over all remaining periods.

2. Permanent income increase of 1 in every remaining period

$$dW_t = \sum_{j=0}^{N-1} \frac{1}{(1+r)^j} = \frac{1}{\theta_N} \quad \Rightarrow \quad \boxed{dC_t = 1}$$

Now lifetime wealth rises by the full annuity value, so current consumption rises one-for-one.

What we learn: consumption reacts strongly to persistent income news and weakly to transitory income news. As $N \rightarrow \infty$, the temporary MPC approaches $\theta_N \rightarrow r/(1+r) \approx 0.03$ at a 3 % real interest rate: a DKK 10,000 windfall raises annual consumption by only DKK 300.

Why This Resolves the Empirical Puzzle

The PIH matches both facts from the beginning of the lecture.

- ▶ **Cross-section:** a household with unusually high income this year often has a **temporary** positive income shock. Consumption rises, but by less than income, so the APC is lower.
- ▶ **Time series:** when the whole economy becomes richer over time, permanent income rises alongside actual income. Consumption then rises roughly proportionally, so the aggregate APC stays stable.

Main lesson: the Keynesian function is too static. The PIH adds the missing intertemporal logic.

Mid-Lecture Check: Think Before We Continue

Go to **Socrative** → Student login → Room **MACROECONOMICSB**.

Question: The Danish government announces a tax rebate of DKK 5,000 per household, paid out immediately. It credibly commits to raising taxes by exactly the same present-value amount next year. Assume perfect capital markets and rational, forward-looking households. By how much should your current consumption rise?

- (A) By DKK 5,000. Spend it all now.
- (B) By roughly DKK 150. Spread the windfall over a lifetime at $r \approx 3\%$.
- (C) By DKK 0. Save everything to pay the future tax bill.
- (D) It depends on whether you can borrow and on your liquid savings.

Submit your answer in Socrative. We will discuss the result after the next section.

The Government's Intertemporal Budget Constraint

The government also faces an intertemporal budget constraint. Let G_1, G_2 be public spending and D_1 the initial debt stock. **Fiscal sustainability** requires:

$$D_1 + G_1 + \frac{G_2}{1+r} = T_1 + \frac{T_2}{1+r}.$$

- ▶ For *given* G_1, G_2 : a tax cut today ($dT_1 < 0$) must be offset by higher taxes tomorrow: $dT_2 = -(1+r)dT_1 > 0$.

Now recall human wealth:

$$H_1 = (Y_1^L - T_1) + \frac{Y_2^L - T_2}{1+r}.$$

Question: what happens to H_1 when the government cuts T_1 and raises T_2 by $(1+r)|dT_1|$?

Ricardian Equivalence

Substituting the government's intertemporal constraint into human wealth:

$$dH_1 = -dT_1 - \frac{dT_2}{1+r} = -dT_1 - \frac{-(1+r)dT_1}{1+r} = -dT_1 + dT_1 = 0.$$

Human wealth is unchanged. Since $C_1 = \theta(V_1 + H_1)$:

$$dC_1 = \theta dH_1 = 0$$

Ricardian Equivalence Theorem (Ricardo 1820; Barro 1974): *A switch from tax finance to debt finance of government spending, with no change in current or planned future spending, leaves private consumption unchanged.*

Households save the entire tax cut to pay the future tax bill. The implied savings rate would have risen sharply in the Socratic example, with no consumption response.

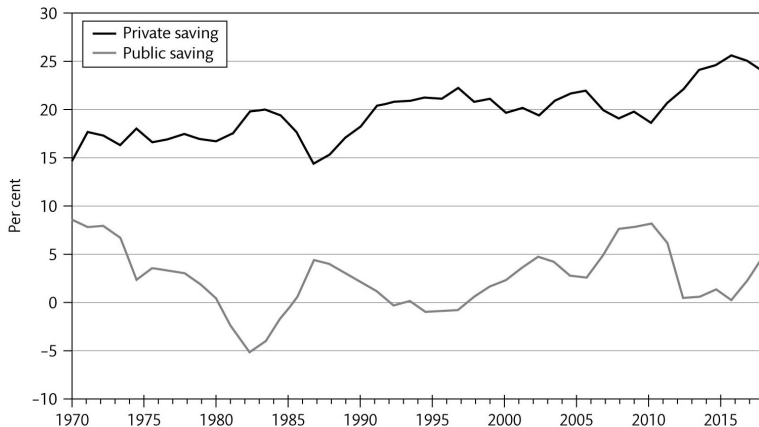
Algebra

Why Ricardian Equivalence Is Likely to Fail

The theorem requires assumptions that rarely hold completely. Three main reasons it breaks down in practice:

1. **Finite horizons and intergenerational redistribution.** Today's taxpayers may not expect to pay all future taxes: part of the burden will fall on generations not yet born. A debt-financed tax cut then raises current households' lifetime resources, lifting consumption.
2. **Credit constraints.** Liquidity-constrained households cannot borrow against future income. A tax cut today relaxes their constraint and raises current consumption immediately, even if they know future taxes will be higher.
3. **Distortionary taxes.** The theorem assumes lump-sum taxes. In practice, income taxes affect labour supply and savings incentives. A temporary cut in income taxes can shift labour supply inter-temporally and affect output, breaking the neutrality result.

Danish Evidence: Private and Public Saving



Source: Sørensen & Whitta-Jacobsen, Figure 16.8. Private and public saving in Denmark, per cent of GDP, 1970–2018 (Danish Economic Council; SMEC database).

Private and public saving often move in opposite directions, but far from one-for-one.

Borrowing Constraints

Suppose the household faces a **borrowing limit**: $A_1 \geq 0$ (cannot borrow at all). Adding this constraint (with Lagrange multiplier $\mu_1 \geq 0$) changes the Euler equation to:

$$u'(C_1) = \mu_1 + \frac{1+r}{1+\phi} u'(C_2), \quad \mu_1 \geq 0$$

- ▶ Constraint **not binding**: $\mu_1 = 0$ and we recover the standard Keynes-Ramsey rule.
- ▶ Constraint **binds**: $\mu_1 > 0$, so $u'(C_1)$ is higher than the unconstrained optimum implies. Consumption today is *lower* than the household would prefer.

Implication: a cash transfer to a constrained household raises current consumption almost one-for-one ($MPC \approx 1$), since the household was spending less than it wanted to. This breaks Ricardian equivalence and explains why observed MPCs can far exceed the PIH benchmark.

Derivation

Precautionary Saving

The basic idea is simple: households dislike very low consumption especially strongly. So uncertainty about future income can change current spending.

Without uncertainty, households try to smooth consumption over time:

$$C_1 \approx C_2.$$

With uncertainty, future income may be high or low:

$$Y_2 = \begin{cases} Y_2^{\text{high}} & \text{good state,} \\ Y_2^{\text{low}} & \text{bad state.} \end{cases}$$

The bad state matters a lot because consumption is then especially valuable. Households therefore save more today to protect themselves against that bad outcome.

$\text{higher future income risk} \Rightarrow C_1 \downarrow, \quad S_1 \uparrow$

Precautionary saving means that households build a buffer stock because they are uncertain about the future.

- ▶ Households may cut spending before income actually falls.
- ▶ The effect is stronger when job risk, refinancing risk, or income risk rises.

The Generalised Consumption Function

Putting the pieces together, consumption depends on both **lifetime resources** and **frictions that make current cash flow matter**. This gives the generalised consumption function:

$$C_t = C\left(Y_t^d, (Y_t^d)^e, r, V_t\right)$$

- ▶ $\partial C / \partial Y_t^d$ is **much larger** in reality than the unconstrained PIH predicts: credit-constrained consumers have a high MPC.
- ▶ Expected future income $(Y_t^d)^e$ matters for unconstrained households through the PIH channel.
- ▶ The interest rate likely has a net *negative* effect once wealth effects through asset prices and discount rates are included.
- ▶ V_t captures housing wealth, stock portfolios, and pension savings.

Back to the Opening Question: Why Adjustable-Rate Borrowers Cut More

The consumption model gives three reasons:

- ▶ $r \uparrow$ lowers current consumption when the substitution effect and wealth effect dominate.
- ▶ Lower house prices and higher discount rates reduce V_t^+ , pulling down consumption through total wealth.
- ▶ If borrowing constraints or tight cash flow matter, the MPC out of disposable income becomes large, so payment resets bite immediately.

A common monetary shock can therefore generate **heterogeneous consumption responses** across households.

Question: Which channel should be strongest for a young homeowner with a large adjustable-rate mortgage and little liquid wealth?

Notebook: L3_consumption.ipynb

- ▶ Solve the two-period consumption problem and verify the closed-form rule $C_1 = \theta(V_1 + H_1)$.
- ▶ Compare the MPC out of temporary and permanent income changes.
- ▶ Plot consumption for constrained and unconstrained households and identify where the borrowing constraint binds.
- ▶ Connect the model to data on consumption and disposable income.

Takeaways

- ▶ The **Keynesian function** $C = \bar{C} + bY^d$ fits cross-sections but predicts a falling APC that is not observed in long-run aggregate data.
- ▶ The **IBC** says: $C_1 + C_2/(1+r) = V_1 + H_1$. Consumption depends on total wealth, not just current income.
- ▶ The **Keynes-Ramsey rule** determines the consumption path: $u'(C_1) = \frac{1+r}{1+\phi} u'(C_2)$.
- ▶ Under the **PIH**, $MPC \approx r/(1+r) \approx 0.03$ for a temporary income gain; a permanent gain shifts consumption proportionally.
- ▶ **Ricardian Equivalence**: a debt-financed tax cut is fully saved by rational households. It fails with credit constraints, finite horizons, or distortionary taxes.
- ▶ **Borrowing Constraints** raise the MPC to near 1 for constrained households; **precautionary saving** reduces C_1 when income risk rises.
- ▶ The **generalised consumption function**: $C = C(Y_+^d, (Y^d)_+^e, r_{(-)}, V_+)$ nests all mechanisms.

Preview: Lecture 4

Lecture 4 combines the consumption function derived today with the investment function from Chapter 15 into a theory of **aggregate demand** (Chapter 17).

- ▶ We will derive the **IS curve**: how does the real interest rate affect aggregate output through both investment and consumption simultaneously?
- ▶ We will connect IS to a monetary policy rule and the Phillips curve.
- ▶ The wealth effects and interest rate channels in the consumption function will be central building blocks.

Appendix

Historical Note: Keynes vs. Kuznets

The empirical puzzle was first identified in the 1940s:

- ▶ **Keynes (1936):** the APC should decline as income rises. This prediction is supported by cross-section microeconomic evidence.
- ▶ **Kuznets (1946):** US time-series data showed a remarkably stable aggregate APC over several decades, despite large income growth. This directly contradicted the Keynesian prediction.
- ▶ **Resolution: Modigliani and Brumberg (1954) and Friedman (1957):** consumption depends on lifetime resources, not current income alone. Cross-sectionally, high current income often reflects a temporarily good year, so the APC appears low. Over the long run, permanent income grows with actual income, so the aggregate APC is stable.

Ricardian Equivalence: The Algebra

Human wealth: $H_1 = (Y_1^L - T_1) + (Y_2^L - T_2)/(1 + r)$.

A debt-financed tax cut satisfies $dT_2 = -(1 + r) dT_1$, so:

$$dH_1 = -dT_1 - \frac{dT_2}{1 + r} = -dT_1 - \frac{-(1 + r) dT_1}{1 + r} = -dT_1 + dT_1 = 0.$$

Since θ and V_1 are unaffected, $dC_1 = \theta \cdot dH_1 = 0$.

What the household does instead: period-1 saving rises by $|dT_1|$, since

$dA_1 = d(V_1 + Y_1^d) - dC_1 = |dT_1| - 0 = |dT_1|$. In period 2, the accumulated assets $(1 + r)|dT_1|$ exactly finance the higher tax bill $dT_2 = (1 + r)|dT_1|$. The household's consumption in period 2 is also unchanged. $\square$

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Precautionary Saving: Jensen's Inequality

Setup: $u'''(C) > 0$, so $u'(\cdot)$ is a *convex* function.

Jensen's inequality for a convex function f and random variable X : $\mathbb{E}[f(X)] > f(\mathbb{E}[X])$.

Applied to $u'(C_2)$:

$$\mathbb{E}[u'(C_2)] > u'(\mathbb{E}[C_2]).$$

Suppose (for contradiction) that $C_1 = \mathbb{E}[C_2]$. The Euler equation requires $u'(C_1) = \frac{1+r}{1+\phi} \mathbb{E}[u'(C_2)]$.

Substituting:

$$u'(\mathbb{E}[C_2]) = \frac{1+r}{1+\phi} \mathbb{E}[u'(C_2)] > \frac{1+r}{1+\phi} u'(\mathbb{E}[C_2]).$$

This requires $\frac{1+r}{1+\phi} < 1$, i.e. $r < \phi$. For $r \geq \phi$, the equality cannot hold.

To restore it, $u'(C_1)$ must be *higher*, meaning C_1 must be *lower*: $C_1 < \mathbb{E}[C_2]$. Uncertainty depresses current consumption below the certainty equivalent. $\square$

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Appendix: Borrowing Constraint Euler Equation

Choose end-of-period assets A_1 to maximise

$$u(C_1) + \frac{1}{1+\phi} u(C_2),$$

subject to

$$C_1 = V_1 + Y_1^d - A_1, \quad C_2 = (1+r)A_1 + Y_2^d, \quad A_1 \geq 0.$$

Set up the Lagrangian:

$$\mathcal{L} = u(V_1 + Y_1^d - A_1) + \frac{1}{1+\phi} u((1+r)A_1 + Y_2^d) + \mu_1 A_1,$$

where $\mu_1 \geq 0$ is the multiplier on the borrowing constraint.

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Appendix: Borrowing Constraint Euler Equation, continued

Differentiate with respect to A_1 :

$$\frac{\partial \mathcal{L}}{\partial A_1} = -u'(C_1) + \frac{1+r}{1+\phi} u'(C_2) + \mu_1 = 0.$$

Rearranging gives

$$u'(C_1) = \mu_1 + \frac{1+r}{1+\phi} u'(C_2).$$

If the constraint does not bind, $\mu_1 = 0$ and you recover the standard Keynes-Ramsey rule. If it binds, $\mu_1 > 0$ and current consumption is below the unconstrained optimum.

$$\mu_1 A_1 = 0 \quad (\text{complementary slackness}).$$

